

## FinTrust Digital Bank – Week 4

### Assumptions, Limitations & Lessons Learned

#### *Data Analytics Track*

**Purpose.** This document records the main assumptions, limitations, responsible-use considerations, lessons learned and final improvement priorities for the FinTrust Data Analytics solution. It supports the Week 4 finalisation stage by making clear what the solution can demonstrate, what it cannot demonstrate, and what was learned while developing, testing and refining the work across Weeks 1–4.

#### 1. Final Assumptions

- **Synthetic project environment:** FinTrust is a fictional organisation and the customer and transaction datasets are synthetic. The analysis is intended for educational and portfolio purposes.
- **Customer and transaction identifiers:** Customer\_ID and Transaction\_ID are treated as the valid unique identifiers for customers and transactions. Customer names are not treated as unique identifiers.
- **Cleaned datasets are the analytical source of truth:** The final SQL, Python and Power BI work uses the cleaned FinTrust customer and transaction datasets prepared during the project. Previously identified missing values in Device\_Type and Location were handled during data preparation.
- **Transaction period:** The available transaction data represents the January–March 2026 period. Findings about trends are therefore limited to this three-month window.
- **Risk review interpretation:** Risk\_Review\_Flag is treated only as an educational review indicator. It is not interpreted as proof of fraud, financial crime or customer wrongdoing.
- **Association, not causation:** Observed relationships, such as higher risk review rates associated with higher transaction value, are interpreted as associations. The analysis does not establish causal relationships.

#### 2. Final Limitations

- **Synthetic data:** Because the data is synthetic, the findings should not be treated as evidence of actual customer behaviour or real banking risk.
- **Short time horizon:** Only three months of transaction history are available. This limits seasonal analysis, long-term trend detection and forecasting.
- **No causal modelling:** The project is descriptive and diagnostic business intelligence work. It does not prove why a pattern occurred or whether one variable caused another.
- **Risk indicator scope:** The Risk\_Review\_Flag is not a fraud label. The dashboard and recommendations therefore refer to review activity rather than confirmed fraud.
- **Limited external context:** The analysis does not incorporate macroeconomic conditions, service outages, campaign calendars, customer complaints, branch events or other external factors that could explain changes in transaction behaviour.
- **Operational recommendations require business validation:** Recommendations are evidence-based within the dataset, but real implementation would require confirmation from operations, risk, product, compliance and technology teams.
- **No production deployment:** The Power BI dashboard, SQL scripts and Python notebooks are analytical project outputs and should not be represented as a production banking system.
- **Cross-track integration:** The Data Analytics work explains how analytics could contribute to the wider FinTrust solution, but no integration with a live Data Science model, ML service or GenAI assistant should be claimed unless those outputs were actually received and used.

### 3. Responsible-Use Considerations

- Do not present Risk\_Review\_Flag as evidence that a customer committed fraud or financial crime.
- Do not use the analysis to make unsupported personalised financial, legal or investment decisions.
- Use customer-segment and engagement findings for analytical understanding, not for unfair or discriminatory treatment.
- Treat high failure rates and high review rates as signals for further investigation, not as proof of root cause.
- If the solution were adapted for real banking use, governance, privacy, security, access control, auditability and regulatory review would be required.

### 4. What Worked Well

1. **Cross-tool validation:** Using SQL, Python and Power BI to check the same findings improved confidence in the results. Monthly trends, risk-review amount bands and failure-rate patterns were reproduced consistently across tools.
2. **Progressive project development:** The work built logically from Week 1 planning to Week 2 descriptive analysis, Week 3 deeper analysis and Week 4 final testing/refinement rather than restarting the project each week.
3. **Decision-oriented dashboard development:** The Power BI dashboard evolved from summary reporting into a more useful BI solution with normalized KPIs, risk-review analysis, a channel × transaction-type failure matrix, customer engagement analysis and customer-segment drill-through.

### 5. Challenges Encountered

1. **Moving beyond descriptive analysis:** The Week 2 analysis was strong descriptively, but Week 3 required deeper comparisons and normalized measures. This was addressed by adding customer-level, month-over-month, amount-band and multi-dimensional analyses.
2. **Maintaining consistency across tools:** The same KPI or analytical result needed to match in SQL, Python and Power BI. This required repeated validation and careful use of consistent definitions and filters.
3. **Power BI usability:** Improving the dashboard without overcrowding it required reorganising visuals, replacing weaker descriptive charts and testing navigation behavior.

### 6. Key Lessons Learned

1. **Validation is part of analysis, not an afterthought:** A finding is stronger when it can be independently reproduced. Cross-checking results helped identify which findings were fully supported and which required a more cautious interpretation.
2. **Totals can be misleading without normalization:** The largest customer segment was not automatically the strongest on a per-customer basis. Measures such as transaction value per customer and transactions per customer provided better comparative insight.
3. **Good dashboards require prioritisation:** Adding more visuals does not automatically improve a dashboard. Replacing basic charts with more decision-relevant visuals and preserving clear page structure improved usability.

## 7. Key Improvements Made Across the Project

| Area          | Improvement                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SQL           | Moved from descriptive summaries to CTEs, window functions, rankings, NTILE, month-over-month analysis, normalized segment analysis and multi-dimensional failure/risk queries. |
| Python        | Extended Week 2 EDA with trend analysis, comparative visualisation, segmentation, risk amount bands and cross-tool validation.                                                  |
| Power BI      | Added normalized KPIs, failure-rate matrix, amount-band risk analysis, digital-engagement views, customer-segment analysis and drill-through.                                   |
| Validation    | Recalculated KPIs and key findings, checked data quality and referential integrity, tested Python reproducibility, and compared SQL/Python/Power BI outputs.                    |
| Documentation | Documented findings, evidence, business meaning, recommendations, limitations, responsible-use considerations and refinement evidence.                                          |

## 8. Recommendations for Future Project Work

- Use a longer time period to support seasonality, trend stability and forecasting.
- Add operational context such as outages, campaign dates, complaint data or service incidents to help explain observed changes.
- Define a formal data dictionary and KPI catalogue early so that SQL, Python and Power BI calculations remain aligned throughout the project.
- Automate more validation checks so data quality and KPI consistency can be rerun quickly after changes.
- For a real deployment, involve risk, compliance, security, operations and product stakeholders before acting on analytical recommendations.

## 9. Final Reflection

The FinTrust project demonstrated a complete analytics workflow from business understanding and data preparation through advanced analysis, dashboard development, testing, validation and final communication. **The strongest improvement across the project was the shift from reporting what happened to validating patterns and explaining what they mean for decision-making.** The final solution remains an educational business intelligence project built on synthetic data, but it demonstrates a structured and reproducible approach to analytical problem-solving.